

CYBERSECURITY ANALYST INTERNSHIP PROGRAM

Task 1: Cybersecurity Fundamentals & Practical Awareness

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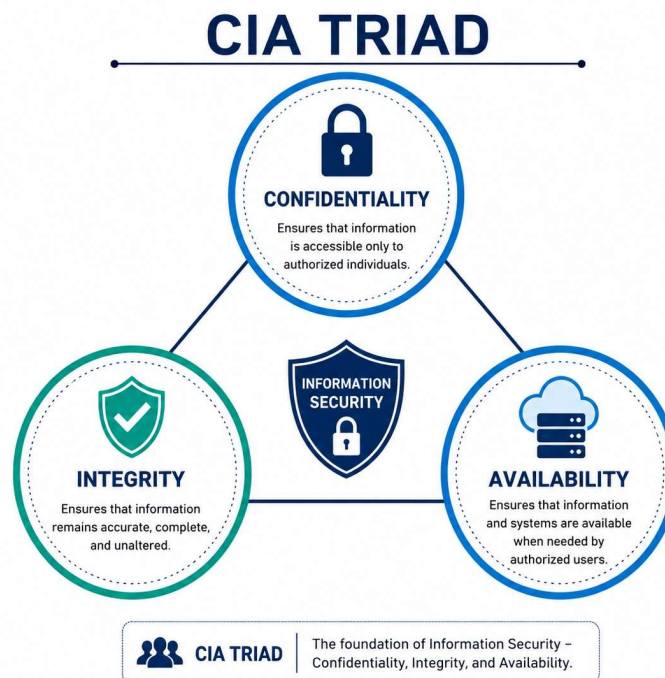
Introduction:

Cybersecurity is the practice of protecting computer systems, networks, applications, and digital information from cyber threats and unauthorized access. As technology continues to evolve, cyberattacks have become more frequent and sophisticated, making cybersecurity an essential part of every organization. The primary objective of cybersecurity is to ensure the confidentiality, integrity, and availability of information.

This report provides an overview of the fundamental concepts of cybersecurity, including the CIA Triad, common cyberattacks, and a real-world cyberattack case study. Understanding these concepts helps build a strong foundation for a career in cybersecurity.

CIA Triad

The **CIA Triad** is the foundation of information security. It consists of three key principles that help organizations protect their information and systems.



1. Confidentiality:

Confidentiality ensures that sensitive information is accessible only to authorized users. Unauthorized individuals should not be able to view or access confidential data.

Examples:

- Email passwords
- ATM PIN codes
- Patient medical records

Methods to Protect Confidentiality:

- Strong passwords
- Multi-Factor Authentication (MFA)
- Data encryption
- Access control

2. Integrity:

Integrity ensures that data remains accurate, complete, and unaltered. Unauthorized users should not be able to modify information.

Examples:

- Student examination results
- Bank transaction records
- Government databases

Methods to Protect Integrity:

- Hashing
- Digital signatures
- File integrity monitoring
- Access permissions

3. Availability:

Availability ensures that systems and information are accessible whenever authorized users need them.

Examples:

- Online banking services
- Hospital management systems
- University portals

Methods to Protect Availability:

- Regular backups
- Firewalls
- Disaster recovery plans
- DDoS protection

Common Cyber Attacks

1. Phishing:

Phishing is a cyberattack in which attackers send fake emails or messages to trick users into revealing passwords, banking details, or other confidential information.

Prevention:

- Verify email senders.
- Avoid clicking suspicious links.
- Enable Multi-Factor Authentication.

2. Malware:

Malware is malicious software designed to damage or gain unauthorized access to computer systems.

Examples:

- Virus
- Worm
- Trojan Horse

Prevention:

- Install antivirus software.
- Keep software updated.
- Download files only from trusted sources.

3. Ransomware:

Ransomware encrypts a victim's files and demands payment to restore access.

Prevention:

- Regular data backups.
- Updated antivirus.
- Do not open unknown email attachments.

4. SQL Injection:

SQL Injection is a web application attack where malicious SQL commands are inserted into input fields to access or manipulate a database.

Prevention:

- Use parameterized queries.
- Validate user input.
- Apply secure coding practices.

5. Cross-Site Scripting (XSS):

XSS occurs when an attacker injects malicious JavaScript into a web page, affecting other users.

Prevention:

- Validate and sanitize user input.
- Encode output.
- Use Content Security Policy (CSP).

6. Brute Force Attack:

A brute force attack involves repeatedly trying different passwords until the correct one is found.

Prevention:

- Use strong passwords.
- Enable account lockout.
- Use Multi-Factor Authentication.

Real-World Cyber Attack Case Study

WannaCry Ransomware Attack (2017):

The WannaCry ransomware attack occurred in May 2017 and affected more than **200,000 computers across over 150 countries**. The attackers exploited a vulnerability in Microsoft Windows known as **EternalBlue**. Once a computer was infected, the ransomware encrypted files and demanded payment in Bitcoin to restore access.

The attack severely disrupted hospitals, businesses, universities, and government organizations. Many healthcare services in the UK experienced delays because their computer systems became unavailable.

Conclusion

Cybersecurity is essential for protecting digital information and maintaining secure systems. The CIA Triad provides the core principles of information security, while understanding common cyberattacks helps organizations defend against modern threats. The WannaCry incident demonstrates the importance of regular updates, strong security practices, and user awareness. Continuous learning and practical experience are key to becoming a successful cybersecurity professional.

References

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2. Microsoft Security Documentation.
3. OWASP Foundation – OWASP Top 10.
4. Cisco Networking Academy – Introduction to Cybersecurity.